Two exact projector matrices

EXACT COMPUTATION / rational entries in the stated S_3 basis

$E_{\langle(12)\rangle}$

	e	(12)	(23)	(13)	(123)	(132)
e	$\frac{1}{2}$	$\frac{1}{2}$				
(12)	$\frac{1}{2}$	$\frac{1}{2}$				
(23)			$\frac{1}{2}$		$\frac{1}{2}$	
(13)				$\frac{1}{2}$		$\frac{1}{2}$
(123)			$\frac{1}{2}$		$\frac{1}{2}$	
(132)				$\frac{1}{2}$		$\frac{1}{2}$

$E_{\langle(23)\rangle}$

	e	(12)	(23)	(13)	(123)	(132)
e	$\frac{1}{2}$		$\frac{1}{2}$			
(12)		$\frac{1}{2}$				$\frac{1}{2}$
(23)	$\frac{1}{2}$		$\frac{1}{2}$			
(13)				$\frac{1}{2}$	$\frac{1}{2}$	
(123)				$\frac{1}{2}$	$\frac{1}{2}$	
(132)		$\frac{1}{2}$				$\frac{1}{2}$

$$(E_H E_K - E_K E_H)\delta_e = \frac{\delta_{(123)} - \delta_{(132)}}{4}$$

Nonzero commutator. Both matrices are still self-adjoint idempotents.